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CONCEPTIONS OF MATHEMATICS AND HOW IT IS LEARNED: THE PERSPECTIVES OF STUDENTS ENTERING UNIVERSITY

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Abstract

This paper reports results from an investigation to identify the conceptions of mathematics held by beginning university students and their approaches to the study of mathematics. Phenomenographic techniques were used to analyze responses to a questionnaire administered to approximately 300 students. An analysis of the results identified a structural relationship between students' conceptions of mathematics and their approaches to learning it, with the majority of students viewing mathematics as a necessary set of rules and procedures to be learned by rote. The results of this research have implications for the ways in which teaching and learning are constituted within universities.

Introduction

Mathematics in school is the foundation for later learning at university level. Research at school level indicates that the learners' previous experiences influence the quality of their approaches to learning, attitudes to and outcomes in learning mathematics (Clarke, 1985; Crawford, 1983, 1986, 1990; Resnick, 1987; Schoenfeld, 1989). There are indications that students' conceptions of mathematics and their orientations to study affect the quality of cognitive activity and of learning outcomes (Crawford, 1986, 1992). Also, the ways in which learners interpret the context of their mathematical learning and hence the ways they relate to mathematical activities influence their mathematical thinking (Cobb, Yackel, & Wood, 1992; Crawford, 1990; Lave, 1988; Solomon, 1989; Steffe & Cobb, 1988). Furthermore, Biggs and Collis (1982) have shown that the characteristics of learning activity as expressed by students' verbal reports about learning reflect differences in thinking in ways which are related to their learning outcomes.

There is also a growing consensus among researchers in the field of student learning in higher education that university students learn from their experiences (Marton, Hounsell, & Entwistle, 1984; Ramsden, 1991). It would be expected that students would

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enter university with a range of conceptions of mathematics and general predispositions to mathematics learning derived from their school activities. There is also evidence that the students' perceptions of the quality of these activities, substantially relates to the ways in which they approach learning at university level and the quality of learning outcomes (Crawford, 1992; Gordon, 1993; Prosser & Millar, 1989; Trigwell & Prosser, 1991a,b; Volet & Lawrence, 1989).

The interest in approaches to learning stems from an acknowledgement that learning is a human activity that occurs in a cultural context. Phenomenographic research, pioneered by the Gothenburg School (Marton, 1988), describes the qualitatively different ways subjects relate to phenomena. This approach views phenomena systemically and avoids the boundaries in traditional psychology between thinking, feeling, and acting — between person and context. As Vygotsky (1978) proposed, there is no assumption of a Cartesian duality between self and context; between thinking and acting. This is in accord with the insistence by activity theorists (e.g., Leont'ev, 1981; Davydov & Markova, 1983) that acting, thinking, and feeling are all part of one coordinated system that interacts in reciprocal ways with known objects. For the activity theorists, the social context for all activity is important and collective external activity precedes individual internal activity. Lave's (1988) definition of a setting as the *relationship* between an actor and the arena in which s/he acts also has some resonance with a phenomenographic approach.

This relational view of learning has formed the basis of much of the research into student learning in higher education. In summary that research indicates that learners' experiences should be considered as involving their orientation in context, the way they relate to the learning environment, their goals or intentions, as well as their learning strategies.

A Phenomenographic Perspective on Learning

A framework used by Marton (Marton & Säljö, 1984; Marton, 1988) to conceptualize learning was to consider it in terms of its referential aspect, or what learning is (a conception of learning) and the structural aspect, or how one goes about it (an approach to learning). This framework describes students' experiences of learning in general and has been used to describe learning in specific disciplines. It has not, as yet, been applied to the learning of mathematics at university level.

Figure 1 (based upon Marton, 1988, p. 66) shows a diagrammatic analysis of the logical relationship between conceptions of and approaches to learning. As is shown, both the "what" of learning (conception) and the "how" of learning (approach) can themselves be recursively analyzed into referential and structural components, that is, analyzed in terms of what is focussed on and how the focus is achieved.

In this framework, the experience of learning can be conceptualized and analyzed in recursive terms. That is, the experience can be conceptualized in terms of a conception of learning and an approach to learning. The conception can be further analyzed in terms of what people focus on when reporting their conception of learning; such as a quantitative increase in the amount of knowledge they have or a changed understanding of reality, and how that focus is achieved; by memorization or by abstraction of meaning.

Similarly, the approach can be analyzed in terms of what is focussed on; a surface

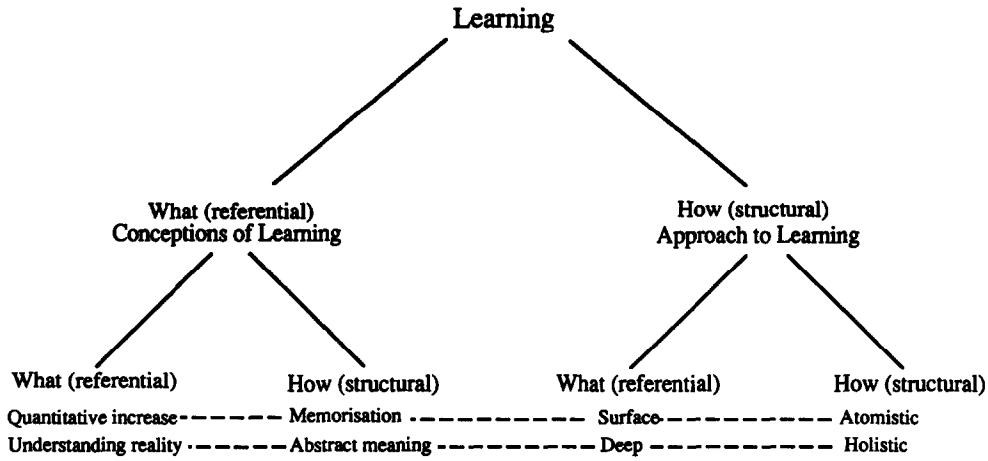

Figure 1. Relationship between conception of learning and approach to learning.

approach versus a deep approach, and how the focus is achieved; an atomistic or a holistic approach.

In surface approaches to learning, the student's attention and activity is centred on reproduction of knowledge that is perceived as transmitted by experts — short term instrumental goals. Self evaluation addresses the question: How can I reproduce their knowledge under assessment conditions? In deep approaches students adopt a more global and personal perspective on learning. Learning strategies are aimed at making sense of new information in terms of existing conceptions and opinions, which may be revised if necessary.

In an atomistic approach to learning, students memorize unconnected routines and pieces of information, while in a holistic approach they concentrate on forming a cohesive view which takes account of the new ideas. Marton (1988) suggests that most students either seek to memorize bits of information in order to reproduce that information or they seek to abstract meaning with the intention of understanding the material they are studying, so that there is a relationship between the referential and structural aspects of approaches to learning.

This phenomenographic perspective provides a basis for the identification of the categories of description of students' experiences of mathematics. Inherent in the approach is a means of analyzing their structure. Thus, this method provides a generic approach to analyzing meanings constituted within particular contexts and content domains.

This study addresses three research questions:

(1) What are the nature and range of 'beginning-university' students' conceptions of mathematics and how are these distributed?
(2) What are the nature and distributions of approaches to learning mathematics for 'beginning-university' students?
(3) Are there relationships between students' initial conceptions of mathematics, their approaches to learning, and their achievements at university level?

Method Of Research

A survey questionnaire on students' conceptions of mathematics and approaches to learning mathematics was administered to approximately 300 first year mathematics students at the University of Sydney. The questionnaire consisted of five open ended questions, two of which are the focus of this paper. The questionnaire was designed to elicit students' own conceptions of mathematics and their approaches to studying it. The theoretical position described above suggests that learning will be shaped by students' perceptions. The closeness of these perceptions to expert opinion is a secondary issue. Therefore, in order to minimize the possibility of "leading" the students' responses, the questions were kept as open and general as possible.

In-depth structured interviews were also carried out with 12 selected students in order to clarify their written statements and to enhance interpretations of the questionnaire responses. The interviews were taped and transcribed.

A phenomenographic approach was taken to analyze the students' responses. The first stage in this analysis was to identify a set of categories of description to the open ended questions. The categories are conceived of as a relationship between the group of researchers and the data. Thus, they are constituted in relation to the awareness of the researchers of recent theories of learning in the field of mathematics education, and the characteristics of the data. The process for reaching agreement on shared meanings involved all members of the research team in the following procedure:

(1) Twenty-one representative survey responses were selected. These were chosen to cover both the range of levels of mathematics studied prior to university and the range of university entrance scores.
(2) Five researchers individually identified an initial set of categories.
(3) Researchers met and compared categories and explored the relationships between them.
(4) The research team agreed to a draft set of categories.
(5) Each researcher took the same twenty-one questionnaires and classified the responses according to the draft categories.
(6) The individual classifications were compared and a final set of clear statements of each category was agreed upon.
(7) One researcher then classified all 300 responses.
(8) Any responses that were difficult to classify were later discussed by all five researchers.
(9) Later, responses involving a mismatch between conceptions of mathematics and approach to learning mathematics were further reviewed by two researchers.

This procedure was followed to categorize the students' conceptions of mathematics and independently to categorize their approaches to learning mathematics.

The students' responses to the questionnaire were then analyzed to explore the relationships between conceptions of mathematics, approaches to learning mathematics and attainment in first year mathematics at university.

Results

In much research, the categories of description or variables are imposed on the data by an expert researcher. In this study, the categories of description are constituted in relation to the data and are a main result of the investigation.

Conceptions of Mathematics

In order to gain insight into beginning university students' conceptions of mathematics, the following question was asked: *Think about the maths you've done so far. What do* you *think mathematics is?*

The analysis of the students' responses to this question resulted in the development of a set of categories of description for their conceptions of mathematics. Brief descriptions of the categories and illustrative quotes are shown in Table 1.

Table 1
Categories of Responses for Conceptions of Mathematics

Category	*Representative quote from student survey*
A. Maths is numbers, rules and formulae	*Maths is the study of numbers, and the application of various methods of changing numbers*
B. Maths is numbers, rules and formulae which can be applied to solve problems	*Mathematics is the study of numbers and their applications in other subjects and the physical world*
C. Maths is a complex logical system; a way of thinking	*Mathematics is the study of logic. Numbers and symbols are used to study life in a systematic perspective and requires the mind to think in a logical and often precise manner*
D. Maths is a complex logical system which can be used to solve complex problems	*Maths is an abstract reasoning process which can be utilized to explore and solve problems*
E. Maths is a complex logical system which can be used to solve complex problems and provides new insights used for understanding the world	*Techniques for thinking about observable, physical phenomena in a quantitative way and also for thinking more abstractly with little or no relation to the directly observable universe*

Note that in Table 1, the statement of category B has not been repeated in the statement of the subsequent categories but its inclusion is assumed.

Although these categories explicitly describe the referential aspects of students' conceptions, implicit within them are the structural aspects. That is, categories A and B present mathematics as a fragmented body of knowledge while categories C, D, and E proffer a cohesive view of mathematics. Table 2 summarizes the logical relationships between the referential and structural aspects of students' conceptions.

In referential terms, category A involves a focus on number rules and formulae. In category B the emphasis is on using these number rules and formulae to solve problems. Category C involves a qualitative shift to seeing mathematics as a complex, logical

Table 2
Relationship among Categories of Description for Conceptions of Mathematics

Referential (what focused on)	Structural (how focused on)	
	Fragmented	Cohesive
(1) Numbers/rules/formulae	A	
(2) As in 1 above and solve problems	B	
(3) As in 2 above and a complex, logical way of thinking		C
(4) As in 3 above and solve complex problems		D
(5) As in 4 above and insights for understanding the world		E

system, while category D highlights the use of this system to solve complex problems. Finally, in category E, the focus goes beyond seeing mathematics in terms of seeing and solving problems, to seeing it as a way of providing insights for understanding the world.

Structurally, categories A and B are fragmented, focusing on parts rather than wholes. Categories C, D, and E, on the other hand, are cohesive, concentrating on the whole picture rather than just the constituent parts.

The categories are both empirically and logically inclusive. They are empirically inclusive since students' reported conceptions of mathematics were found to include responses indicating the conceptions described in preceding categories.

The logically inclusive nature of the relationships between the categories is highlighted in Table 2. It shows the hierarchically nested relationship between the referential aspects of the categories. For example, the referential aspect of category E is inclusive of the referential aspect of category D in the sense that there is an extension in student awareness from D to E. This is, a student who conceives of mathematics as a way of helping us understand the world has a more extended awareness than a student who only conceives of mathematics as a way of thinking that can be used to solve problems. It can also be seen that categories A and B differ from categories C, D, and E in structural terms. While the conceptions cannot be thought of as inclusive in terms of their structure, since cohesive and fragmented views of mathematics are mutually exclusive, the categories are logically inclusive in that they are ordered by their referential aspects.

Student Approaches to Study

The original questionnaire was designed so that students had two opportunities to report their approaches to the study of mathematics prior to university.

They were:

Think about some maths you understood really well. How did you go about studying that? (It may help you to compare how you studied this with something you feel you didn't fully understand.)
How do you usually go about learning some maths?

Consequently, the responses to these questions were analyzed together to identify a set of qualitatively different categories of description for students' approaches to studying mathematics. The analysis resulted in the construction of a logically related and an empirically but not logically inclusive set of categories of description of student approaches. These categories of description and representative student quotes to illustrate them are given in Table 3.

Table 3
Categories of the Responses for Students' Approaches to Learning Mathematics

Category	*Representative quote from student survey*
A. Learning by rote memorization, with an intention to reproduce knowledge and procedures	*I liked calculus because I could remember formulas which is how I used to study. I would rote learn all the formulas and summarize all my theoretical notes*
B. Learning by doing lots of examples, with an intention to reproduce knowledge and procedures	*The way I go about studying for mathematics is by doing a lot of questions and examples. Firstly I would study the notes and learn formulas, then I put all of that to use by doing heaps of exercises*
C. Learning by doing lots of examples with an intention of gaining a relational understanding of the theory and concepts	*To understand a topic well it was important to gain an understanding of the basic concepts involved, backed up by some problem solving on the topic. However, concepts which were not fully comprehended could become well understood through extra work on related questions: i.e., it is essential to do a wide range of questions on a topic to fully understand it*
D. Learning by doing difficult problems, with an intention of gaining a relational understanding of the entire theory, and seeing its relationship with existing knowledge	*After listening to explanation of how a particular maths works the most essential features a repetition to develop speed (this usually consists of boring menial tasks) and an equal component of very difficult problems which require a great deal of thought to explore that area and its various properties and their consequences*
E. Learning with the intention of gaining a relational understanding of the theory and looking for situations where the theory will apply	*Read the relevant theory and try to get on the same "wavelength" as the person who actually discovered it. Before I attempt any problems I try to think where you can use the concept: i.e., what the concept was invented for. Then I attempt problems (on my own)*

Again, the referential and structural aspects of the responses were the basis of the data analysis; these being the students' intentions when studying mathematics, and the strategies used to achieve them. The relationships between the referential and structural aspects of the categories describing students' approaches are shown in Table 4.

In referential terms, the approaches described in categories A and B show an intention to reproduce knowledge and procedures, while those in C, D, and E show an intention to develop a relational understanding of the mathematics; to make sense of new knowledge. Thus, there is a qualitative shift in intention that occurs between categories B and C from

Table 4
Relationships Between the Referential and Structural Aspects of Students' Approaches

	Intention (referential)	
Strategy (structural)	Reproduction	Understanding
Rote memorization	A	
Doing lots of examples	B	C
Doing difficult problems		D
Applying the theory		E

learning for the purpose of reproduction to learning for meaning. In the terms of the Gothenburg school (Marton, 1988) this shift is from a surface to a deep approach.

In considering the structural aspects it can be seen that the approach in category A emphasizes rote memorization as a strategy to learn mathematics, while those in B and C share the more active strategy of doing lots of examples. The approach in D focuses on the strategy of doing more difficult problems to explore the subject so as to gain an understanding, while approach E goes further and looks for situations where the theory can apply.

The approaches identified by these categories are empirically inclusive in the sense that students who report adopting approach D to the study of mathematics would also describe having approaches A, B, and C at their disposal, whereas a student whose approach to the study of mathematics was categorized as A would report on using that approach alone. However, there is no logical inclusion here, since a student who has the intention of studying for understanding cannot simultaneously include an intention of just reproducing knowledge and procedures. Further, any one strategy of learning does not logically incorporate a preceding strategy.

Illustration Of Categories: Two Examples

The major qualitative results of this study were the categories described in the previous sections. In this section these categories are illustrated by referring to the responses of two students. For these students, responses from their surveys were compared with those given to the same questions during interviews which were conducted some time after the surveys were completed. In the interviews with selected students the researchers attempted to clarify students' written responses and flush them out. Efforts were made to investigate whether the interpretations of the survey material were reliable, that is, whether the researchers' categorization of students' survey responses matched that of their interview responses. In fact, the interview material was remarkably consistent with the survey data as exemplified by these excerpts.

Conceptions of Mathematics

The conception of mathematics of the first student, who shall be referred to as Sally, was classified in category B, one of the two categories in which the structure

of mathematics was viewed as fragmented. Sally's conception included an awareness of the preceding concept of mathematics as being about numbers, rules or formulae (category A) but her main focus was on solving problems. To the survey question *What do you think mathematics is?* Sally reported:

> "I feel mathematics is the process of using different techniques to solve various problems. In my experience this has only involved numbers yet I feel it may be helpful for a wider range of future subjects."

In her interview she confirmed this conception of mathematics describing mathematics as:

> "Oh, numbers and problems and getting an actual answer to a problem . . ."

Pressed to expand on this point she added:

> "It is a way of working out problems and I suppose you can use it in the future to relate it to bigger things. I hate this question!"

She, like many other students, felt uncomfortable with the unfamiliar demand to reflect more deeply on this question.

The other side of the shift in conceptions that occurs between the categories A and B and the other three categories C, D, and E, is typified by George. George did not answer the question *What do you think mathematics is?* in his survey but his response to one of the other questions led the researchers to categorize his conception in category C as: *Maths is a complex logical system; a way of thinking.*

He wrote:

> "I am doing maths for the purpose of improving a logical method of thinking rather than for the course specific subject matter . . ."

In the interview he elaborated on this focus:

> "Well more than anything I say that it is probably a thinking process which is not something you get from any other subject and that is probably the reason I am doing it . . . more than anything as a way of thinking. I know that is not a definition of maths but that's what I see as the most important thing behind what I am doing."

Evidence that he perceived the structure of mathematics as cohesive was also given during the interview:

> "Well I think I try to see what is being taught as part of the bigger picture, to see how everything comes into place and I find if I can get a broad sort of overview of things then it probably helps me to understand the work a little better, because I can see where it is heading, where it is going to fall into place. I think someone who is not so good at maths might struggle to see where it is heading, and they're just sort of interested in getting hold of one thing at a time."

Approaches to Learning Mathematics

Some of the diversity in students' awareness of their approaches to learning mathematics is also illustrated by the survey and interview responses of these two students. Sally expressed a strategy for learning mathematics of doing lots of examples for practice. In her survey she wrote:

> "I usually read summaries & try to memorise formulas when necessary. Also answering questions & doing past papers is what I do . . ."

Her interview account of her strategy was consistent with her written report:

> "I used to do past papers and plenty of problems and writing down formulae and stuff. . . . In the hope that I'll be able to recall them in an exam, to go round about the short way of answering something in an exam, maybe. I'm probably just trying to memorize things which, I don't know if that's a good idea but . . ."

Hence, at times Sally was using a category A strategy of rote memorization but her repertoire extended beyond this to include the procedure of doing lots of examples for practice. This strategy and her stated intention that she was trying to reproduce what she had learnt led the researchers to categorize her approach as B: *Learning by doing lots of examples with an intention to reproduce knowledge and procedures.*

George was classified at the other end of the spectrum in category D, that is,

Learning by doing difficult problems, with an intention of gaining a relational understanding of the entire theory, and seeing its relationship with existing knowledge.

His survey response was:

> "Do as many different types of questions as I can. Try to see how the topic being studied is related to other topics I have studied — get the full picture."

This was borne out by the interview response reported above:

> "Well I think I try to see what is being taught as part of the bigger picture, to see how everything comes into place . . ."

He also expressed the view that while approaches representative of preceding categories were adopted at times, these were not his exclusive ways of learning mathematics.

> ". . . Like what I said before when I go through and do my notes first I like to get an understanding of the principles behind the topic. . . . I don't go back time after time and do the same sort of question. When I am revising I go back and do the same sort of thing again. But when I am trying to learn it, then each type of question, I might only do one or two."

The above examples suggest a relationship between students' conceptions of mathematics and their approaches to learning it. This is borne out by the quantitative analyses described below.

Quantitative Analyses

In this section the distribution of results across categories for both the conceptions and approaches are described, as well as the relationship between conceptions and approaches. In addition, it is shown how the conceptions and approaches relate to student performance. Table 5 presents the distribution across categories of conceptions and approaches. In this sample, there was no evidence of gender effect.

Table 5 shows that 77% of conception statements were classified in the first two (fragmented) categories, while 23% were classified in the last three (cohesive) categories. Eighty two percent of the approach statements were classified in one of the two (surface) categories, while only 18% were classified in one of the three (deep) categories.

Table 6 shows a cross-tabulation of the relationship between prior conception and prior approach.

Table 5
Distribution of Prior Conceptions and Approaches

Conception/Approach	*N*	% of responses
Conception		
A. Maths as numbers, rules and formulae	62	26
B. Maths as numbers etc. with applications to problems	124	51
C. Maths as way of thinking	32	13
D. Maths as way of thinking for complex problem solving	18	7
E. Maths provides new insights for understanding the world	6	3
Missing Data (no response)	52	
Total	294	
Approach		
A. Learning by rote memorization for reproduction	17	6
B. Learning by doing examples for reproduction	215	76
C. Learning by doing examples to develop relational understanding	30	11
D. Learning by doing difficult problems to extend understanding	15	5
E. Learning by looking for broader context to extend understanding	6	2
Missing Data (no response)	11	
Total	294	

Table 6
Relationship Between Prior Conception and Prior Approach

Conceptions	Approach		Total
	Surface (A & B)	Deep (C, D, & E)	
Fragmented (A & B)	179 (91%)	17 (9%)	196
Cohesive (C, D, & E)	4 (10%)	36 (90%)	40
Total	183	53	236

Chi-square = 126, $p<.001$

Note that in Table 6, the terms fragmented/cohesive are used to describe the conceptions of mathematics rather than just the structural aspects of the conceptions of mathematics. Similarly, the terms surface/deep are used to describe the approaches to studying mathematics, not just the referential aspects of the approaches to studying mathematics.

Table 6 indicates a relationship between conception and approach. It shows that 91% of the students who had a fragmented view of high school mathematics had adopted a surface approach to the study of high school mathematics, while only 10% of those who had a cohesive conception had adopted a surface approach.

Table 7 indicates that students with a cohesive conception of mathematics and/or able to adopt a deep approach to study tended to achieve at a higher level, after a year of university study. These are medium effect sizes according to Cohen (Cohen, 1977).

Table 7
Relationship Between Prior Conception and Prior Approach and Achievement after a Year of University Study

Conception/Approach		Mathematics 1 Semester 2 exam results	
		‡Mean	SD
Conceptions			
	Fragmented	123	21
	Cohesive	134	23
	T Test: $T=$	2.16*	
Approaches			
	Surface	122	29
	Deep	136	21
	T Test: $T=$	2.66†	

*$p<.05$. † $p<.01$.
Note. ‡Exam result marked out of 200.

Discussion

The discussion of the results of the analysis will address three issues:

(a) The usefulness of the methodology for studying students' experiences of studying mathematics.
(b) What the results say about the nature of students' experiences in studying mathematics.
(c) Issues for teaching and learning mathematics.

A Phenomenographic Perspective on Learning Mathematics

The phenomenographic method has allowed the emphasis of the research to be on the students' own perspectives on mathematics and how it is learned. That is, by keeping the questions as general as possible, care was taken not to shape students' responses. The method used for developing categories of description from the data, rather than imposing predetermined categories based on the personal knowledge and perspectives of the researchers, has improved the ability of the researchers to explore the students' points of view, so enhancing the ethnographic validity of this research. The use of students' reports of conceptions and approaches, although limited by their awareness and abilities in written verbal expression, has provided a rich source of information beyond second order interpretation of student behaviour. The two distinct types of intention associated with the learning strategy of practising many examples (see Table 4) is a particularly interesting result of the kind not available through observation. More to the point, since the result was entirely unexpected for the researchers, it was unlikely to be revealed by imposed categories of description or through usual methods of content analysis. Although a quantitative analysis of the distribution of resulting categories and their relationships to assessment scores at the university has been reported in this paper,

the emphasis of this research has been on the phenomenon of students' conceptions of mathematics and the ways in which they approach learning.

Students' Experiences of Studying Mathematics

Figure 2 summarizes the results of the study. It shows how the experiences identified in the study can be structured in terms of the recursive model described in the Introduction. Students' experiences of learning can be described referentially in terms of the students' conception of what mathematics is and structurally in terms of their approach to learning mathematics. Their conceptions can, in turn, be described referentially in terms of mathematics being about numbers, rules, and formulae to it being about a complex logical system providing insights about the world. Structurally, the focus is achieved in a fragmented or cohesive way. Similarly, their approaches can be described referentially in terms of having an intention to reproduce or understand, and structurally in terms of adopting strategies ranging from rote memorization to looking for new situations where the theory will apply.

This analysis, together with the results in Table 6, highlights one aspect of the essentially relational interpretation of the experience of studying mathematics from a phenomenographic perspective. Table 6 shows the strong empirical relationship between the way students conceive of mathematics and the way they approach their studies of it. This empirical relationship is consistent with the logical analysis shown in Figure 2, and together they suggest that the conception and approach are internally related phenomena. The conception and approach are not constituted independently, but are constituted dynamically in relation to each other. That is, students' conceptions of mathematics are formed by their approaches to learning it and also form their approaches.

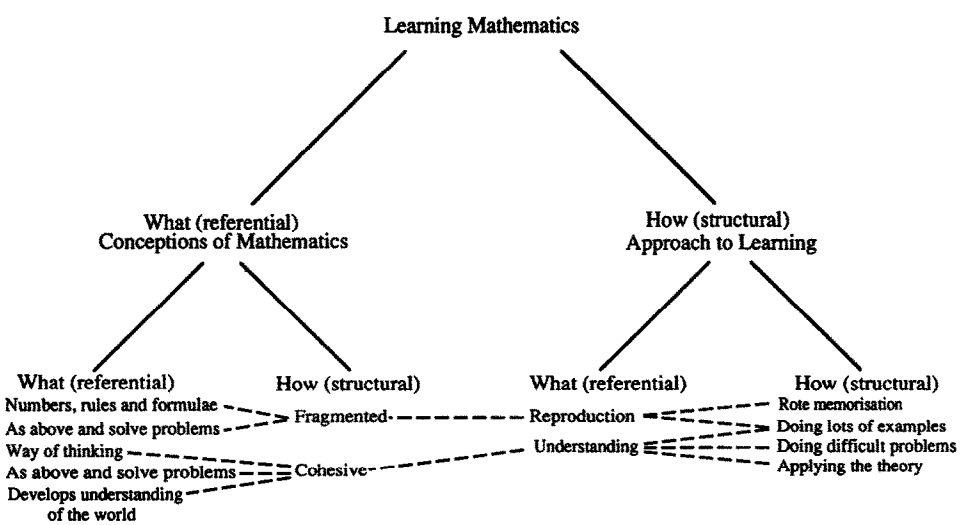

Figure 2. Relationship between conception of mathematics and approach to learning mathematics.

Neither the conception nor the approach should be considered as being prior to the other, or dependent upon the other. Focusing students' attention on either the approach or conception and hoping that changes in one will cause changes in the other is therefore unlikely to be successful.

Issues For Teaching And Learning Mathematics

This research indicates that university teachers need to consider not only the content of their course, but also how the students conceive of what that content is about and how they go about learning it. There is a need to explicitly address the relationship between the student and the learning context. That is, if the students perceive that the workload is too great, or believe that the assessment measures reproduction, then they are more likely to adopt low level conceptions and surface approaches to learning as their best strategy for success.

Students' perceptions of assessment tasks and how these perceptions orient them within the university setting is one major factor that constitutes this relationship between the student and the learning context. For example, the problem is particularly salient when using multiple choice questions (Scouller & Prosser, in press). It is very difficult to convince students that multiple choice questions measure anything other than reproduction, even when the questions are designed to measure understanding.

Another major factor is what constitutes the object of study in first year mathematics courses. The object of study is constituted for the students and the teachers in terms of what is presented and how it is presented (Entwistle & Marton, 1994). The presentation of mathematics at university often takes the form of lectures on a finished and polished product. As Dreyfus (1991) notes, students ". . . have been taught the products of the activity of scores of mathematicians in their final form, but they have not gained insight into the processes that have led mathematicians to create these processes" (p. 28).

Overall, the conditions of learning mathematics at university level too often unintentionally place the student in a position where short term instrumental strategies are, in fact, an expedient means to success.

While this paper has focussed on students' learning as they enter higher education, other research focuses on teachers' conceptions of teaching and learning and approaches to teaching (Prosser, Trigwell, & Taylor, (in press); Trigwell, Prosser, & Taylor, 1994). These studies have shown that a substantial number of teachers in first year university physical science courses hold conceptions of learning that are reproductive in nature. Many physical science teachers in first year university courses believe that reproductive learning is what is appropriate at that level. There is no reason for doubting that a similar situation exists in the teaching of many first year university mathematics courses.

In conclusion, this research emphasizes the need for a shift in attention away from teaching and learning as independent activities towards a more systemic examination of the relationships between the two activities and the context in which they occur.

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